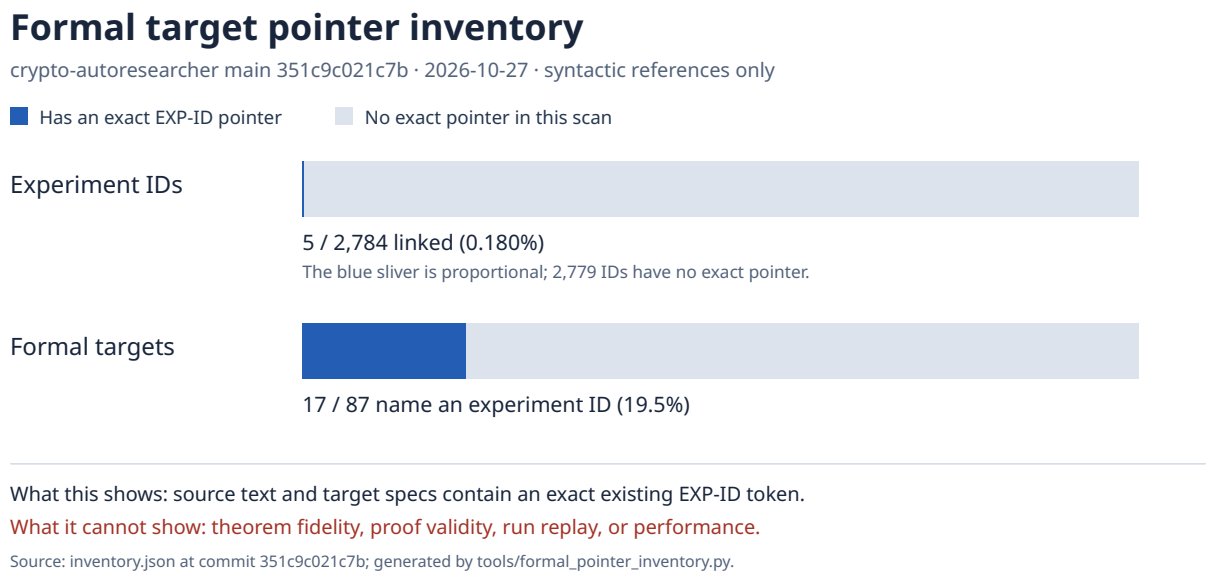
Exact experiment ID pointers in formal targets

crypto-autoresearcher, 2026-10-27; source commit `351c9c021c7be044ade54dc7e352b33d3a772244`

Question. Which existing experiment-directory IDs occur as exact tokens in a registered formal target, its declared text source, or its Lean theorem source? This is a syntactic inventory, not a theorem or run review.



Method. The repository tool reads immutable Git objects, enumerates tracked `EXP-*` directories and all 87 registered target YAMLs, then scans target and named text-source bytes for exact existing IDs. No source exceeded the 1 MB scan limit. The machine-readable `inventory.json` binds the tree hashes, all IDs, and each witness path.

Result. Five of 2,784 experiment IDs have an exact pointer in 17 of 87 registered targets. The five are BINSTD-8bdf86, FROB-91ee9c, FROB-b8cf21, GFPN-05ff43, and NCP-001 (all prefixed `EXP-`). NCP-001 occurs in the declared NCP theory note and was absent from the earlier local diagnostic. Its pointer does not certify the full experiment claim.

Boundary. A present pointer does not establish Lean validity, semantic fidelity, archived execution, or a performance/security claim. An absent pointer does not establish that no relevant theorem exists. This report covers the committed autoresearcher `main` tree only; `crypto`, `cryptanalysis`, open PRs, and uncommitted work need separate inventories. No research decision or status changed.

Reproduction. Run the command below from the repository root. The complete method, limits, and next checks are in `REPORT.md`.

```
python3 tools/formal_pointer_inventory.py \
  --repo . \
  --revision 351c9c021c7be044ade54dc7e352b33d3a772244 \
  --out /tmp/pointers.json
```